

Guidance Document on Tigecycline Dosing

in association with
Revision of Breakpoints for *Enterobacterales*

July 2022

Background

EUCAST revised tigecycline breakpoints in 2021-22 based on more recently published preclinical and clinical data. The revised breakpoints and the considerations that led to them were published in a new version of tigecycline rationale document (v 2.0), which should be read in conjunction with this guidance.

During the revision of breakpoints, EUCAST examined the literature and sought advice on dosing practices of tigecycline. EUCAST wished to determine whether there was frequent use of dosing regimens that gave higher exposures than those in the current licensed regimen.

EMA-approved dosing

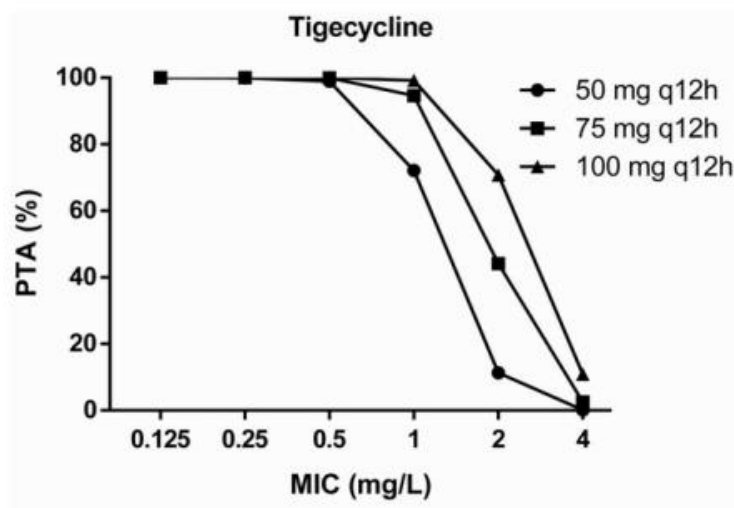
The European Medicines Agency's Summary of Product Characteristics for tigecycline has the following regimens listed:

- Adults and Adolescents aged 12 to <18 years: an initial dose of 100 mg followed by 50 mg every 12 hours
- Children aged 8 to <12 years: 1.2 mg/kg of tigecycline every 12 hours intravenously to a maximum dose of 50 mg every 12 hours

Higher-exposure dosing regimens

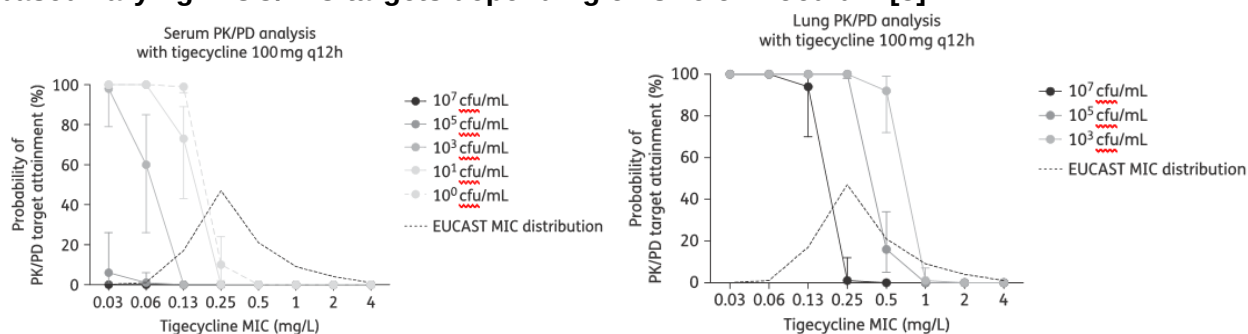
There is some limited published literature on higher-exposure dosing regimens with tigecycline. Ni et al. [1] undertook Monte Carlo simulation analyses of standard and high-dose tigecycline (200 mg loading dose followed by 100 mg every 12 hours) against local isolates of carbapenemase-producing *Klebsiella pneumoniae* and found that the cumulative fractional response rate of more than 90% was only achievable with the high-dose regimen (Figure 1).

Figure 1 PTA at various dosages of tigecycline (Ni et al, 2018) based on an *fAUC/MIC* target of 0.9 identified in the study of Bhavnani et al [2]



Tsala et al. studied the exposure response relationship of a simulated dosage of 50 mg x 2 against strains of *K. pneumoniae* with MICs ranging from 0.125 to 2 mg/L [3]. The $fAUC_{24}/MIC$ values were influenced by the size of the inoculum. Better responses were seen with simulated unbound lung concentrations ($fC_{max} = 0.125$ mg/L) than with serum concentrations ($fC_{max} = 1.25$ mg/L). Probabilities of Target attainment are shown in Figure 2.

Figure 2 PTA of serum versus lung exposures at a dosage of 50 mg x 2 of tigecycline based varying $fAUC/MIC$ targets depending on size of inoculum [3].



Published treatment studies are shown in Table 1.

Table 1 Published studies on higher-exposure tigecycline regimens

Authors [Ref.]	Year	Patient group	Tigecycline regimen	No. of patients receiving >100 mg per day
De Pascale et al. [4]	2014	Critically ill patients	Standard: 50 mg every 12 hours High dose: 100 mg every 12 hours	46 (VAP 33)
Baron et al.[5]	2018	All patients receiving tigecycline	Various	4
Geng et al. [6]	2018	Carbapenemase-producing <i>K. pneumoniae</i> bloodstream infections	200mg loading dose; 100 mg every 12 hours	17
Chen and Shi [7]	2018	MDR pathogens	100 mg every 12 hours	69
			100 mg every 12 hours for 3-5 days, followed by 50 mg every 12 hours	11
Gao et al. [8]	2018	Febrile neutropenia in acute leukaemia	100 mg every 12 hours	41
Han et al., [9]	2021	Secondary bloodstream infection with XDR <i>A. baumannii</i>	Standard: 50 mg every 12 hours versus High dose: 100 mg every 12 hours	Standard dosage n=63 High dosage n=88
Qu et al., [10]	2021	Carbapenem-resistant gram-negative bacteraemia		Standard dosage n=30

Authors [Ref.]	Year	Patient group	Tigecycline regimen	No. of patients receiving >100 mg per day
				High dosage n=22
Bai et al., [11]	2021	Pneumonia due to MDR- <i>A. baumannii</i>	Standard: 50 mg every 12 hours and High dose: 100 mg every 12 hours	

These studies reported favourable efficacy, and no excess rates of adverse reactions compared to the standard dosing regimen, either directly within the study or compared to published rates. In the study by De Pascale et al. [2], although there was no difference in cure rate ($p=0.5$), mortality ($p=0.14$) or microbiological eradication ($p=0.07$) in the high dose group in the ventilator-associated pneumonia subset using univariate analysis, multivariable logistic regression showed that high dose tigecycline was an independent predictor of clinical cure (odds ratio 6.25, CI: 1.59-24.57).

High dose tigecycline was found to be superior to standard dose tigecycline in patients with a range of serious *Acinetobacter baumannii* infections complicated by secondary bacteraemia except when the underlying infection was pulmonary [8]. Other investigators have not been able to show a benefit of high dose compared to standard dose in carbapenem-resistant gram-negative bacteraemia more generally [9] or *A. baumannii* pneumonia [10]. In these studies, tigecycline was mostly used in combination.

A recent systematic review and meta-analysis ten studies comparing high versus standard dose tigecycline demonstrated 30-day all-cause mortality, clinical cure and microbiological eradication benefit for high versus standard dosage overall. However, the authors noted the risk of bias was high because most studies were observational, involved predominantly combination therapy and had small sample sizes [12]. An earlier meta-analysis of high dose tigecycline reached similar conclusions, but noted that these differences were confined to Chinese patients [13]. Both meta-analyses concluded that high dose did not increase the rate of adverse reactions.

Summary

Evidence for the efficacy and safety of high-dose tigecycline, most commonly 100 mg 12-hourly with or without a 200 mg loading dose, is generally positive but limited and potentially biased.

Recommendations

High-dose tigecycline may be considered in seriously ill patients infected with pathogens resistant to all other classes of antimicrobials. The PK-PD of tigecycline at these high doses predicts that MDR-strains with tigecycline MICs of up to 1 mg/L will respond to treatment. This cut-off applies to the wild type of some species of *Enterobacterales* other than *E. coli*, most notably *K. pneumoniae* and *K. oxytoca*. Species with ECOFFs of 4 or 8 mg/L, such as *Proteus* and *Providencia* spp. are considered intrinsically resistant.

High dose tigecycline may also be considered in non-pulmonary XDR-*Acinetobacter baumannii* infections.

Recommendations for high-dose regimens have been published by Cunha et al. [14].

References

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